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Reactor apparatus

Abstract

A reactor apparatus, includes: a reactor chamber having an inlet through which treatment liquid containing by-products is introduced and having an interior space; a burner at a lower end portion of the reactor chamber to burn waste gas; a guide member above the burner and configured to allow the treatment liquid to flow outwardly of the burner; a water reservoir between the burner and the guide member, the water reservoir having a double pipe structure having an inner wall portion and an outer wall portion, and through which water supplied through a water inlet is configured to flow between the inner wall portion and the outer wall portion; and a cover member coupled to an upper end portion of the water reservoir and configured to cover a space between the inner wall portion and the outer wall portion, wherein an upper end of the outer wall portion is above an upper end of the inner wall portion, wherein a plurality of bumps are on a bottom surface of the cover member spaced apart from each other in a circumferential direction, the plurality of bumps configured to form a gap of several hundred μm between the bottom surface of the cover member and an upper surface of the inner wall portion of the water reservoir.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION(S)

(1) This application claims benefit of priority to Korean Patent Application No. 10-2021-0087146 filed on Jul. 2, 2021 in the Korean Intellectual Property Office, the disclosure of which is incorporated by reference herein in its entirety.

BACKGROUND

(2) The present inventive concepts relate to a reactor apparatus.

(3) A scrubber facility is used to treat toxic waste gas emitted from a facility after a semiconductor manufacturing process.

(4) Among scrubber facilities, a burn-wet scrubber facility having a structure in which waste gas is first combusted in a combustion chamber in the scrubber and a by-product is accommodated therein using a treatment liquid is used.

SUMMARY

(5) An aspect of the present inventive concepts is to provide a reactor apparatus capable of compensating for powder emissions, which may be a by-product of combustion, and in some cases, may cause operation of an entire semiconductor manufacturing facility to be stopped.

(6) An aspect of the present inventive concepts is to provide a reactor apparatus capable of forming a more uniform water film on an inner surface of an inner wall portion of a water reservoir.

(7) According to an aspect of the present inventive concepts, a reactor apparatus, includes: a reactor chamber having an inlet through which treatment liquid containing by-products is introduced and having an interior space; a burner at a lower end portion of the reactor chamber to burn waste gas; a guide member above the burner and configured to allow the treatment liquid to flow outwardly of the burner; a water reservoir between the burner and the guide member, the water reservoir having a double pipe structure having an inner wall portion and an outer wall portion, and through which water supplied through a water inlet is configured to flow between the inner wall portion and the outer wall portion; and a cover member coupled to an upper end portion of the water reservoir and configured to cover a space between the inner wall portion and the outer wall portion, wherein an upper end of the outer wall portion is above an upper end of the inner wall portion, and a plurality of bumps are on a bottom surface of the cover member spaced apart from each other in a circumferential direction, the plurality of bumps configured to form a gap of several hundred μm being between the bottom surface of the cover member and an upper surface of the inner wall portion of the water reservoir.

(8) According to an aspect of the present inventive concepts, a reactor apparatus, includes: a reactor chamber having an inlet through which treatment liquid containing by-products is introduced and having an interior space; a burner at a lower end portion of the reactor chamber configured to burn waste gas; a guide member above the burner and configured to allow the treatment liquid to flow outwardly of the burner; a water reservoir between the burner and the guide member and having a double pipe structure having an inner wall portion and an outer wall portion, and through which water supplied through a water inlet is configured to flow between the inner wall portion and the outer wall portion and is configured to flow along an upper surface and an inner surface of the inner wall portion; and a cover member coupled to an upper end portion of the water reservoir configured to provide a flow path of the water together with the inner wall portion, wherein the upper surface of the inner wall portion includes a plurality of bumps, spaced apart from each other in a circumferential direction, the plurality of bumps configured to form a gap of several hundred μm between a bottom surface of the cover member and the upper surface of the inner wall portion of the water reservoir, wherein the cover member includes a guide block configured to guide a flow path of water passing through the gap on the bottom surface, and a stepped groove on an outside of the bump in a radial direction on the bottom surface.

(9) According to an aspect of the present inventive concepts, a reactor apparatus, includes: a water reservoir between a burner and a guide member and having a double pipe structure having an inner wall portion and an outer wall portion, and through which water supplied through a water inlet is configured to flow between the inner wall portion and the outer wall portion and is configured to flow along an upper surface and an inner surface of the inner wall portion; and a cover member coupled to an upper end portion of the water reservoir configured to provide a flow path of the water together with the inner wall portion, wherein the upper surface of the inner wall portion includes a plurality of bumps, spaced apart from each other in a circumferential direction, the plurality of bumps configured to form a gap of several hundred μm between a bottom surface of the

cover member and the upper surface of the inner wall portion of the water reservoir, wherein the cover member includes a guide block configured to guide a flow path of water passing through the gap on the bottom surface, and a stepped groove on an outside of the bump in a radial direction on the bottom surface.

Description

BRIEF DESCRIPTION OF DRAWINGS

- (1) The above and other aspects, features, and advantages of the present inventive concepts will be more clearly understood from the following detailed description, taken in conjunction with the accompanying drawings, in which:
- (2) FIG. 1 is an illustrative diagram schematically illustrating a scrubber facility including a reactor apparatus according to example embodiments of the present inventive concepts;
- (3) FIG. 2 is a schematic configuration diagram illustrating a reactor apparatus according to example embodiments of the present inventive concepts;
- (4) FIG. 3 is a bottom view illustrating a cover member provided in a reactor apparatus according to example embodiments of the present inventive concepts;
- (5) FIG. 4 is an enlarged view illustrating portion A of FIGS. 2 and 3;
- (6) FIG. 5 is a bottom view illustrating a cover member provided in a reactor apparatus according to example embodiments of the present inventive concepts;
- (7) FIG. 6 is a schematic cross-sectional view illustrating a portion of a cover member and a water reservoir provided in a reactor apparatus according to example embodiments of the present inventive concepts;
- (8) FIG. 7 is a schematic cross-sectional view illustrating a portion of a cover member and a water reservoir provided in a reactor apparatus according to example embodiments of the present inventive concepts;
- (9) FIG. 8 is a schematic cross-sectional view illustrating a portion of a cover member and a water reservoir provided in a reactor apparatus according to example embodiments of the present inventive concepts;
- (10) FIG. 9 is a bottom view illustrating a cover member provided in a reactor apparatus according to example embodiments of the present inventive concepts;
- (11) FIG. 10 is a schematic configuration diagram illustrating a reactor apparatus according to example embodiments of the present inventive concepts;
- (12) FIG. 11 is a plan view illustrating a cover member of a reactor apparatus according to example embodiments of the present inventive concepts; and
- (13) FIG. 12 is an enlarged view illustrating portion B of FIG. 10.

DETAILED DESCRIPTION

- (14) Hereinafter, preferred example embodiments of the present inventive concepts will be described with reference to the accompanying drawings.
- (15) First, a scrubber facility will be described briefly.
- (16) FIG. 1 is an illustrative diagram schematically illustrating a scrubber facility including a reactor apparatus according to example embodiments of the present inventive concepts.
- (17) Referring to FIG. 1, waste gas flowing into a scrubber facility 10 passes through a reactor 12 in which a burner 11 is installed, passes through a wet absorption unit 13, and then is discharged externally. The waste gas flowing into the reactor 12 is combusted by the burner 11 to primarily remove harmful substances contained in the waste gas, and then the waste gas flowing into the wet adsorption unit 13 is accommodated in treatment liquid sprayed through a nozzle 13a, so that harmful substances are secondarily removed. Thereafter, harmful substances are removed once again while passing through an adsorbent 14 containing an absorbing material, and waste gas

passing through the adsorbent is accommodated in the treatment liquid sprayed through the nozzle **13a** so that the harmful substances are finally removed.

(18) The treatment liquid may be provided to the wet adsorption unit **13** after passing through a heat exchanger **15**.

(19) As described above, the gas from which harmful substances are removed is finally discharged externally of the scrubber facility **10**.

(20) FIG. **2** is a schematic configuration diagram illustrating a reactor apparatus according to example embodiments of the present inventive concepts.

(21) Referring to FIG. **2**, a reactor apparatus **100** may include a reactor chamber **110**, a burner **120**, a guide member **130**, a reservoir **140** (for example, a water reservoir) and/or a cover member **150**.

(22) The reactor chamber **110** has an interior space. In addition, the reactor chamber **110** is formed to extend from an upper surface thereof, and has an inlet **112** through which treatment liquid containing by-products is introduced from the wet adsorption unit **13** (please refer to FIG. **1**) described above. The reactor chamber **110** has a shape in which the treatment liquid flowing into the inlet **112** flows to a lower end portion and is discharged externally. For example, the reactor chamber **110** may have a funnel shape with an upper end portion wider than a lower end portion. In addition, an outlet **114** is provided in the reactor chamber **110** so that the treatment liquid and water supplied from the water reservoir **140** can be discharged externally. As an example, the outlet **114** may be disposed below the burner **120**. As described above, since the outlet **114** is provided in the reactor chamber **110**, the treatment liquid, water, and by-products flowing into the reactor chamber **110** may be discharged externally of the reactor chamber **110**.

(23) In addition, a waste gas supply pipe **116** through which waste gas is supplied to the burner **120** may be connected to a bottom surface of the reactor chamber **110**.

(24) The burner **120** is disposed at a lower end portion of the reactor chamber **110**, and is connected to a waste gas supply pipe **116**. In addition, the burner **120** is disposed below the guide member **130**, and is disposed below a hollow of the water reservoir **140**. Accordingly, it is possible to reduce or prevent the treatment liquid flowing in from the inlet **112** from falling toward the burner **120**. The waste gas supplied from a lower portion of the burner **120** is combusted in the burner **120**, and thus, harmful substances contained in the waste gas are primarily removed.

(25) The guide member **130** is disposed above the burner **120**, and serves to allow the treatment liquid to flow externally of the burner **120**. As an example, the guide member **130** is disposed below the inlet **112** of the reactor chamber **110**, and allows the treatment liquid to flow externally of the water reservoir **140**. To this end, the guide member **130** may have a size sufficient to cover the entire water reservoir **140**. As an example, the guide member **130** may have a rounded shape such that an edge thereof is disposed below a central portion thereof. Accordingly, the treatment liquid introduced through the inlet **112** may flow along the inner surface of the reactor chamber **110**. The guide member **130** may have a smaller size than an upper end portion of the reactor chamber **110** so that the waste gas can flow to the wet adsorption unit **13** (please refer to FIG. **1**) disposed thereon.

(26) A water reservoir **140** is disposed between the burner **120** and the guide member **130**. The water reservoir **140** has a double-tube structure having an inner wall portion **142** and an outer wall portion **144**. The water reservoir **140** may include a water inlet **146** through which water flowing through the inner wall portion **142** and the outer wall portion **144** is supplied. As an example, the water supplied through the water inlet **146** flows upwardly through a space between the inner wall portion **142** and the outer wall portion **144**, and then flows down along the inner surface of the inner wall portion **142**. To this end, the upper end of the outer wall portion **144** is disposed above the upper end of the inner wall portion **142**. Accordingly, water flowing upwardly through the space between the inner wall portion **142** and the outer wall portion **144** may flow toward the inner surface of the inner wall portion **142**.

(27) The cover member **150** is coupled to an upper end portion of the water reservoir **140**, and is disposed to cover the space between the inner wall portion **142** and the outer wall portion **144**.

(28) FIG. 3 is a bottom view illustrating a cover member provided in a reactor apparatus according to an example embodiments of the present inventive concepts and FIG. 4 is an enlarged view showing portion A of FIGS. 2 and 3 according to example embodiments of the present inventive concepts.

(29) Referring to FIGS. 3 and 4, a cover member **150** has a hollow shape, and, for example, may have a circular ring shape. In addition, bumps **152** are provided on a bottom surface of the cover member **150**. A plurality of bumps **152** may be disposed to be spaced apart from each other in a circumferential direction, and three bumps **152** may be provided on the bottom surface of the cover member **150** as an example. The bump **152** serves to form a gap of several hundred μm between the bottom surface of the cover member **150** and an upper surface of the inner wall portion **142** of the water reservoir **140**. As described above, by forming a gap S of several hundred μm between the bottom surface of the cover member **150** and the upper surface of the inner wall portion **142** of the water reservoir **140** by the bump **152**, water filled in the water reservoir **140** may flow through the gap S between the bottom surface of the cover member **150** and the upper surface of the inner wall portion **142** of the water reservoir **140** by water pressure. Accordingly, while maintaining a constant flow velocity, water flows to the inner surface of the inner wall portion **142** of the water reservoir **140** so that a water film formed on the inner surface of the inner wall portion **142** of the water reservoir **140** may be more uniformly disposed. In other words, it is possible to reduce or prevent a water film from being formed on the inner surface of the inner wall portion **142** of the water reservoir **140**.

(30) A height of the bump **152** may be determined by the following Equation. That is, when a flow velocity of water at the inlet **146** of the water reservoir **140** is $V1$, an area of the inlet **146** is $A1$, a flow rate thereof is $Q1$, a flow velocity in a gap between the bottom surface of the cover member **150** and the upper surfaces of the inner wall portion **142** of the water reservoir **140** is $V2$, and an area of a gap between the bottom surface of the cover member **150** and the upper surface of the inner wall portion **142** of the water reservoir **140**, that is, an area of the outlet is $A2$, and a flow rate thereof is $Q2$, the height of the bump **152** may be determined to satisfy $Q1(=V1 \times A1) > Q2(=V2 \times A2)$.

(31) A guide block **154** may be provided on the bottom surface of the cover member **150** for guiding a flow path of the water passing through the gap S. The guide block **154** is disposed inside the bump **152** in a radial direction. As an example, the guide block **154** may be formed at the inner end of the bottom surface of the cover member **150** to have a ring shape when viewed from the bottom. The guide block **154** serves to flow along the inner surface of the inner wall portion **142** of the water reservoir **140**, the water passing through the gap S. To this end, the guide block **154** may be formed as a protrusion formed to protrude downwardly.

(32) In addition, a stepped groove **156** disposed outside of the bump **152** in a radial direction may be provided on the bottom surface of the cover member **150**. The stepped groove **156** serves to generate a pressure difference so that water can flow more smoothly through the gap S. As an example, the stepped groove **156** may have a circular band shape when viewed from below the cover member **150**. As an example, the stepped groove **156** is disposed above the upper surface of the inner wall portion **142** and is disposed below the upper surface of the outer wall portion **144**.

(33) Here, when the term for the direction is defined, a radial direction refers to a direction from an outer circumferential surface of the cover member **150** toward a center of a hollow, and a circumferential direction refers to a direction of rotation along an outer circumferential surface of the cover member **150**.

(34) A sealing member **170** is installed between an outer circumferential surface of the cover member **150** and the outer wall portion **144** of the water reservoir **140**. To this end, an installation groove **158** into which the sealing member **170** is inserted may be provided on the outer circumferential surface of the cover member **150**. However, in some example embodiments, a case in which the installation groove **158** is formed in the cover member **150** is described as an example,

but it is not limited thereto, and the installation groove **158** may be formed on the outer wall portion **144** of the water reservoir **140**.

(35) An inclined surface **160** for guiding a flow of waste gas may be provided on an inner circumferential surface of the cover member **150**. The inclined surface **160** serves to allow the waste gas to flow while spreading outwardly of the cover member **150** in a radial direction. Accordingly, the waste gas may flow outwardly of the guide member **130** disposed above the reactor chamber **110** to flow upwardly more smoothly.

(36) As described above, a more uniform water film may be formed on the inner surface of the inner wall portion **142** of the water reservoir **140** through the cover member **150**. Accordingly, it is possible to reduce or prevent powder from accumulating, and it is possible to reduce or prevent powder containing a large amount of moisture from flowing down toward the burner **120** as the powder is accumulated, so that it is possible to reduce or prevent the powder containing a large amount of moisture from extinguishing the burner.

(37) FIG. 5 is a bottom view illustrating a cover member provided in a reactor apparatus according to example embodiments of the present inventive concepts.

(38) Referring to FIG. 5, a cover member **250** has a hollow shape, and may have a circular ring shape as an example. In addition, bumps **252** are provided on a bottom surface of the cover member **250**. A plurality of bumps **252** may be disposed to be spaced apart from each other in a circumferential direction, and as an example, six bumps **252** may be provided on the bottom surface of the cover member **250**.

(39) A guide block **254** disposed inside of the bump **252** in a radial direction may be provided on the bottom surface of the cover member **250**, and the guide block **254** may have a circular ring shape when viewed from below.

(40) In addition, a stepped groove **256** disposed on an outside of the bump **252** in a radial direction may be provided on the bottom surface of the cover member **250**.

(41) FIG. 6 is a schematic cross-sectional view illustrating a portion of a cover member and a water reservoir provided in a reactor apparatus according to example embodiments of the present inventive concepts.

(42) Referring to FIG. 6, a cover member **350** and a water reservoir **340** may be joined by welding. As an example, a joint portion **302** joined by welding may be formed on an outer circumferential surface of the cover member **350** and an inner surface of the outer wall portion **344** of the water reservoir **340**. As described above, it is possible to reduce or prevent water flowing into a space between the inner wall portion **342** and the outer wall portion **344** from flowing toward an outer surface of the outer wall portion **344** by the joint portion **302**. Furthermore, the end of the outer wall portion **344** may be disposed above the end of the inner wall portion **342**. Accordingly, it is possible to reduce or prevent water from flowing toward the outer surface of the outer wall portion **344**.

(43) The cover member **350** may be provided with a bump (not shown), a guide block **354**, a stepped groove **356**, and/or an inclined surface **360**, that are the same or substantially the same configurations as those of the cover member **150** described above. Accordingly, detailed descriptions of the bump (not shown), the guide block **354**, the stepped groove **356** and/or the inclined surface **360** will be omitted.

(44) FIG. 7 is a schematic cross-sectional view illustrating a portion of a cover member and a water reservoir provided in a reactor apparatus according to example embodiments of the present inventive concepts.

(45) Referring to FIG. 7, a cover member **450** is provided with an installation groove **458** into which a sealing member **470** is inserted. The sealing member **470** is installed in the installation groove **458** so as to be in contact with an outer surface of the outer wall portion **444** of the water reservoir **440**. As described above, by increasing a size of the cover member **450**, the cover member **450** may be installed in the water reservoir **440** so that an upper end portion of the outer

wall portion **444** is coupled to a coupling groove **462** of the cover member **450**. In some example embodiments, the installation groove **458** may be connected to the coupling groove **462**.

(46) The cover member **450** may be provided with a bump (not shown), a guide block **454**, a stepped groove **456**, and/or an inclined surface **460**, that are the same or substantially the same configurations as those of the cover member **150** described above. Accordingly, detailed descriptions of the bump (not shown), the guide block **454**, the stepped groove **456** and/or the inclined surface **460** will be omitted.

(47) FIG. **8** is a schematic cross-sectional view illustrating a portion of a cover member and a water reservoir provided in a reactor apparatus according to example embodiments of the present inventive concepts.

(48) Referring to FIG. **8**, a cover member **550** and a water reservoir **540** may be joined by welding. As an example, a joint portion **502** joined by welding may be formed on one surface of a coupling groove **562** of the cover member **550** and an outer surface of the outer wall portion **544** of the water reservoir **540**. As described above, water flowing into a space between the inner wall portion **542** and the outer wall portion **544** by the joint portion **502** can be reduced or prevented from flowing toward a lower portion of the outer surface of the outer wall portion **544**. Furthermore, the end of the outer wall portion **544** may be disposed above the end of the inner wall portion **542**.

Accordingly, it is possible to reduce or prevent water from flowing toward the outer surface of the outer wall portion **544**.

(49) The cover member **550** may be provided with a bump (not shown), a guide block **554**, a stepped groove **556**, and/or an inclined surface **560**, having the same or substantially the same configurations as those of the cover member **150** described above. Accordingly, detailed descriptions of the bump (not shown), the guide block **554**, the stepped groove **556** and/or the inclined surface **560** will be omitted.

(50) FIG. **9** is a bottom view illustrating a cover member provided in a reactor apparatus according to example embodiments of the present inventive concepts.

(51) Referring to FIG. **9**, a cover member **650** has a hollow shape, and may have a circular ring shape as an example. In addition, a bump **652** is provided on a bottom surface of the cover member **650**. The plurality of bumps **652** may be disposed to be spaced apart from each other in a circumferential direction, and three bumps **652** may be provided on the bottom surface of the cover member **650** as an example.

(52) A guide block **654** disposed inside the bump **652** in a radial direction may be provided on the bottom surface of the cover member **650**, and the guide block **654** may have a circular ring shape when viewed from below.

(53) In addition, a stepped groove **656** disposed outside of the bump **652** in a radial direction may be provided on the bottom surface of the cover member **650**. As an example, a plurality of stepped grooves **656** may be disposed to be spaced apart from each other in a circumferential direction.

(54) FIG. **10** is a schematic block diagram illustrating a reactor apparatus according to example embodiments of the present inventive concepts, FIG. **11** is a plan view illustrating a cover member of a reactor apparatus according to example embodiments of the present inventive concepts, and FIG. **12** is an enlarged view illustrating portion B of FIG. **10**.

(55) Referring to FIGS. **10** to **12**, a reactor apparatus **700** may include a reactor chamber **710**, a burner **720**, a guide member **730**, a water reservoir **740**, and/or a cover member **750**.

(56) The reactor chamber **710** has an interior space. In addition, the reactor chamber **710** is formed to extend from an upper surface thereof, and has an inlet **712** through which treatment liquid containing by-products is introduced from the above-described wet adsorption unit **13** (see FIG. **1**). The reactor chamber **710** has a shape in which the treatment liquid introduced through the inlet **712** flows to a lower end portion and is discharged externally. For example, the reactor chamber **710** may have a funnel shape having an upper end portion wider than a lower end portion. In addition, the reactor chamber **710** is provided with an outlet **114** so that the treatment liquid and water

supplied from the water reservoir **740** can be discharged externally. The outlet **714** may be disposed below the burner **720**, for example. As described above, since the outlet **714** is provided in the reactor chamber **710**, treatment liquid, water, and by-products, flowing into the reactor chamber **710**, may be discharged externally of the reactor chamber **710**.

(57) The outlet **714** may be disposed below the burner **720**, for example. As described above, since the outlet **714** is provided in the reactor chamber **710**, a treatment liquid, water, and by-products, flowing into the reactor chamber **710**, may be discharged externally of the reactor chamber **710**. In addition, a waste gas supply pipe **716** through which waste gas is supplied toward the burner **720** may be connected to a bottom surface of the reactor chamber **710**.

(58) The burner **720** is disposed at a lower end portion of the reactor chamber **710**, and is connected to the waste gas supply pipe **716**. In addition, the burner **720** is disposed below the guide member **730**, and disposed below a hollow of the water reservoir **740**. Accordingly, it is possible to reduce or prevent the treatment liquid flowing through the inlet **712** from falling toward the burner **720**. The water gas supplied from a lower portion of the burner **720** is combusted through the burner **720**, and accordingly, harmful substances contained in the waste gas are primarily removed.

(59) The guide member **730** is disposed above the burner **720**, and serves to allow the treatment liquid to flow outwardly of the burner **720**. As an example, the guide member **730** is disposed below the inlet **712** of the reactor chamber **720**, and allows the treatment liquid to flow outwardly of the water reservoir **740**. To this end, the guide member **730** may have a size to cover the entire water reservoir **740**. The guide member **720** may have a rounded shape such that, for example, an edge thereof is disposed to be lower than a central portion thereof. Accordingly, treatment liquid flowing into the inlet **712** may flow along an inner surface of the reactor chamber **710**. In addition, the guide member **730** may have a smaller size than an upper end portion of the reactor chamber **710** so that the waste gas may flow to the wet adsorption unit **13** (please refer to FIG. 1) disposed thereabove.

(60) The water reservoir **740** is disposed between the burner **720** and the guide member **730**. The water reservoir **740** has a double-tube structure having an inner wall portion **742** and an outer wall portion **744**. The water reservoir **740** may include a water inlet **746** through which water flowing between the inner wall portion **742** and the outer wall portion **744** is supplied. As an example, the water supplied through the water inlet **746** flows upwardly through the space between the inner wall portion **742** and the outer wall portion **744**, and then flows downwardly along the inner surface of the inner wall portion **742**. To this end, an upper end of the outer wall portion **744** is disposed above an upper end of the inner wall portion **742**. Accordingly, water flowing upwardly through the space between the inner wall portion **742** and the outer wall portion **744** may flow toward the inner surface of the inner wall portion **742**.

(61) As shown in FIG. 11, bumps **742a** are provided on an upper surface of the inner wall portion **742**. A plurality of bumps **742a** are disposed to be spaced apart from each other in a circumferential direction, and three bumps **742a** may be provided on the upper surface of the inner wall portion **742** as an example. The bump **742a** serves to form a gap of several hundred μm between the bottom surface of the cover member **750** and the upper surface of the inner wall **742** of the water reservoir **740**. As described above, by forming a gap **S** of several hundred μm between the bottom surface of the cover member **750** and the upper surface of the inner wall portion **742** of the water reservoir **740** by the bumps **742a**, water filled in the water reservoir **740** may flow through the gap **S** between the bottom surface of the cover member **750** and the upper surface of the inner wall portion **742** of the water reservoir **740** through water pressure. Accordingly, while maintaining a constant flow velocity, water flows to the inner surface of the inner wall portion **742** of the water reservoir **740** so that a water film formed on the inner surface of the inner wall portion **742** of the water reservoir **740** may be more uniformly disposed. In other words, it is possible to reduce or prevent a water film from being formed on the inner surface of the inner wall portion **742** of the water reservoir **740**.

(62) A height of the bump **742a** may be determined by the follow Equation. That is, when a flow velocity of water at the inlet **746** of the water reservoir **740** is **V1**, an area of the inlet **746** is **A1**, a flow rate thereof is **Q1**, a flow velocity in a gap between the bottom surface of the cover member **750** and the upper surfaces of the inner wall portion **742** of the water reservoir **740** is **V2**, and an area of a gap between the bottom surface of the cover member **750** and the upper surface of the inner wall portion **742** of the water reservoir **740**, that is, an area of the outlet is **A2**, and a flow rate thereof is **Q2**, the height of the bump **742a** may be determined to satisfy $Q1(=V1 \times A1) > Q2(=V2 \times A2)$.

(63) The cover member **750** is coupled to an upper end portion of the water reservoir **740**, and is disposed to cover a space between the outer wall portion **744** and the inner wall portion **742**. As an example, the cover member **750** may have a hollow shape, and as an example, may have a circular ring shape.

(64) As shown in more detail in FIG. **12**, a guide block **754** for guiding a flow of water passing through the gap **S** may be provided on a bottom surface of the cover member **750**. The guide block **754** is disposed radically inside the bump on **742a**. As an example, the guide block **754** may be formed at an inner end of the bottom surface of the cover member **750** to have a ring shape when viewed from the bottom. The guide block **754** serves to flow the water passing through the gap **S** along the inner surface of the inner wall portion **742** of the water reservoir **740**. To this end, the guide block **754** may be formed of a protrusion formed to protrude downwardly.

(65) In addition, a stepped groove **756** disposed externally on the bump **742a** in a radial direction may be provided on a bottom surface of the cover member **750**. The stepped groove **756** serves to generate a pressure difference so that water can flow more smoothly through the gap **S**. As an example, the stepped groove **756** may have a circular band shape when viewed from below the cover member **750**. As an example, the stepped groove **756** is disposed above the upper surface of the inner wall portion **742** and is disposed below the upper surface of the outer wall portion **744**.

(66) A sealing member **770** is installed between an outer circumferential surface of the cover member **750** and the outer wall portion **744** of the water reservoir **740**. To this end, an installation groove **758** into which the sealing member **770** is inserted may be provided on the outer circumferential surface of the cover member **750**. However, in the present example embodiments, a case in which the installation groove **758** is formed in the cover member **750** is described as an example, but it is not limited thereto, and the installation groove **758** may also be formed in the outer wall portion **744** of the water reservoir **740**.

(67) In addition, an inclined surface **760** for guiding a flow of waste gas may be provided on an inner circumferential surface of the cover member **750**. The inclined surface **760** serves to allow the waste gas to flow while spreading outwardly of the cover member **750** in a radial direction. Accordingly, the waste gas may flow outwardly of the guide member **730** disposed above the reactor chamber **710** to flow upwardly more smoothly.

(68) As described above, a more uniform water film may be formed on the inner surface of the inner wall portion **752** of the water reservoir **740** through the cover member **750**. Accordingly, it is possible to reduce or prevent powder from being stacked, and it is possible to reduce or prevent the powder containing a large amount of moisture from flowing down toward the burner **720** as the powder is accumulated, so that it is possible to reduce or prevent powder containing a large amount of moisture from extinguishing the burner.

(69) As set forth above, according to the present inventive concepts, a reactor apparatus capable of forming a more uniform water film on an inner surface of an inner wall portion of a water reservoir may be provided.

(70) The various and advantageous advantages and effects of the present inventive concepts are not limited to the above description, and can be more easily understood in the course of describing specific example embodiments of the present inventive concepts.

(71) While example embodiments have been illustrated and described above, it will be apparent to

those skilled in the art that modifications and variations could be made without departing from the scope of the present inventive concepts as defined by the appended claims.

Claims

1. A reactor apparatus, comprising: a reactor chamber having an inlet through which treatment liquid containing by-products is introduced and having an interior space; a burner at a lower end portion of the reactor chamber configured to burn waste gas; a guide member above the burner and configured to allow the treatment liquid to flow outwardly of the burner; a water reservoir between the burner and the guide member, the water reservoir having a double pipe structure having an inner wall portion and an outer wall portion, and through which water supplied through a water inlet is configured to flow between the inner wall portion and the outer wall portion; and a cover member coupled to an upper end portion of the water reservoir and configured to cover a space between the inner wall portion and the outer wall portion, wherein an upper end of the outer wall portion is above an upper end of the inner wall portion, wherein a bottom surface of the cover member includes a plurality of bumps spaced apart from each other in a circumferential direction, the plurality of bumps configured to form a gap of several hundred μm between the bottom surface of the cover member and an upper surface of the inner wall portion of the water reservoir.
2. The reactor apparatus of claim 1, wherein the plurality of bumps are on the bottom surface of the cover member in contact with the inner wall portion.
3. The reactor apparatus of claim 2, wherein the cover member further comprises a guide block on a bottom surface of the guide member, configured to guide a flow path of water passing through the gap.
4. The reactor apparatus of claim 3, wherein the cover member further comprises a stepped groove on the bottom surface of the cover member on an outside of at least one of the plurality of bumps in a radial direction.
5. The reactor apparatus of claim 4, wherein the stepped groove is above the upper surface of the inner wall portion and below an upper surface of the outer wall portion.
6. The reactor apparatus of claim 1, wherein the cover member further comprises a sealing member between an outer circumferential surface of the cover member and the outer wall portion.
7. The reactor apparatus of claim 1, wherein the cover member has a circular ring shape having a hollow.
8. The reactor apparatus of claim 1, wherein an inner circumferential surface of the cover member has an inclined surface configured to guide a flow of waste gas.
9. The reactor apparatus of claim 1, wherein the cover member and the outer wall portion are joined by welding.
10. The reactor apparatus of claim 1, wherein the cover member further comprises a sealing member between an outer surface of the outer wall portion and a coupling groove of the cover member.
11. The reactor apparatus of claim 1, wherein the reactor chamber further comprises an outlet through which the treatment liquid introduced into the inlet and the water supplied to the water reservoir are discharged, the outlet being below the burner.
12. The reactor apparatus of claim 11, wherein the reactor chamber includes a funnel shape having an upper end portion thereof wider than a lower end portion.
13. A reactor apparatus, comprising: a reactor chamber having an inlet through which treatment liquid containing by-products is introduced and having an interior space; a burner at a lower end portion of the reactor chamber configured to burn waste gas; a guide member above the burner and configured to allow the treatment liquid to flow outwardly of the burner; a water reservoir between the burner and the guide member and having a double pipe structure having an inner wall portion and an outer wall portion, and through which water supplied through a water inlet is configured to

flow between the inner wall portion and the outer wall portion and is configured to flow along an upper surface of the inner wall portion and an inner surface of the inner wall portion; and a cover member coupled to an upper end portion of the water reservoir configured to provide a flow path of the water together with the inner wall portion, wherein the upper surface of the inner wall portion includes a plurality of bumps, spaced apart from each other in a circumferential direction, the plurality of bumps configured to form a gap of several hundred μm between a bottom surface of the cover member and the upper surface of the inner wall portion of the water reservoir, wherein the cover member includes a guide block configured to guide a flow path of water passing through the gap on the bottom surface of the cover member, and a stepped groove on an outside of at least one of the plurality of bumps in a radial direction on the bottom surface of the cover member.

14. The reactor apparatus of claim 13, wherein the plurality of bumps are provided on the upper surface of the inner wall portion and configured to contact the bottom surface of the cover member.

15. The reactor apparatus of claim 14, wherein an upper end of the outer wall portion is above an upper end of the inner wall portion.

16. The reactor apparatus of claim 13, wherein the cover member further comprises a sealing member between an outer circumferential surface of the cover member and the outer wall portion.

17. The reactor apparatus of claim 13, wherein the stepped groove is above the upper surface of the inner wall portion and is below an upper surface of the outer wall portion.

18. A reactor apparatus, comprising: a water reservoir between a burner and a guide member and having a double pipe structure having an inner wall portion and an outer wall portion, and through which water supplied through a water inlet is configured to flow between the inner wall portion and the outer wall portion and is configured to flow along an upper surface of the inner wall portion and an inner surface of the inner wall portion; and a cover member coupled to an upper end portion of the water reservoir configured to provide a flow path of the water together with the inner wall portion, wherein the upper surface of the inner wall portion includes a plurality of bumps, spaced apart from each other in a circumferential direction, the plurality of bumps configured to form a gap of several hundred μm between a bottom surface of the cover member and the upper surface of the inner wall portion of the water reservoir, wherein the cover member includes a guide block configured to guide a flow path of water passing through the gap on the bottom surface of the cover member, and a stepped groove on an outside of at least one of the plurality of bumps in a radial direction on the bottom surface of the cover member.

19. The reactor apparatus of claim 18, wherein an inner circumferential surface of the cover member has an inclined surface configured to guide a flow of waste gas.

20. The reactor apparatus of claim 18, wherein the cover member further comprises a sealing member between an outer surface of the outer wall portion and a coupling groove of the cover member.
